Tip-Guided Few-Shot Adaptation of Motion Time-Series Foundation Models

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Abstract

Cross-dataset few-shot human activity recognition (HAR) requires a pretrained motion encoder to classify activities on an unseen sensor dataset given only k labeled windows per class. Motion foundation models such as UniMTS supply strong zero-shot text alignment, yet two standard adaptation routes both fall short: full fine-tuning overfits sparse supports, while frozen Tip-Adapter caches saturate once domain shift exceeds what a fixed encoder can absorb. We introduce **Tip-Guided Adaptation (TipGuided)**, which fine-tunes the UniMTS ST-GCN encoder under an explicit Tip-cache prior that preserves CLIP-aligned text scores and support affinity structure. On the official UniMTS protocol—18 real HAR datasets, $k \in \{1, 2, 3, 5, 10\}$, leave-one-dataset-out Tip hyper-parameters—TipGuided raises mean accuracy by up to $+8.4$ points over frozen Tip-F and surpasses full fine-tuning at every k, with a peak margin of $+3.8$ points at $k = 2$. Training dynamics show TipGuided enters with higher test accuracy from Tip/text priors and fits the support objective faster; t-SNE embeddings show more class-separable target features after adaptation.

1 Introduction

Wearable IMU sensing powers healthcare monitoring, sports analytics, and mobile context awareness. Recent motion time-series foundation models, notably UniMTS [1], contrastively align a skeleton ST-GCN encoder with CLIP [3] text embeddings and report competitive zero-shot transfer across heterogeneous HAR corpora. Deployment on a new dataset still benefits from a handful of labeled windows per class. This defines a precise and practical problem: **cross-dataset few-shot HAR**.

Two adaptation families dominate this setting. Full fine-tuning updates the encoder and a linear head on the support set; it is strong, but prior-free optimization on $k \leq 10$ windows per class readily memorizes sensor-specific artifacts. Frozen alternatives keep the encoder fixed and learn a lightweight decision rule—linear probes, nearest neighbors, or Tip-Adapter [2] feature caches imported from CLIP few-shot classification. On UniMTS features, Tip-Adapter-F clearly outperforms probes and 1-NN. Further frozen refinements (soft values, late linear fusion, longer key training) add only ~ 0.5 points on average. The bottleneck is therefore not cache hyper-parameter search: **domain shift demands encoder updates that retain pretrained text–motion geometry**.

We propose **Tip-Guided Adaptation (TipGuided)**. TipGuided freezes the CLIP text tower and updates the ST-GCN encoder together with Tip keys and a CE head. Classification logits fuse three complementary signals (Figure 1):

$$z(x) = z_{\text{text}}(x) + \alpha\, A(x)V + \gamma\, W f_\theta(x), \tag{1}$$

where z_{text} is the zero-shot text score, $A(x)V$ is Tip affinity over support keys/values, and $W f_\theta(x)$ is a CE head on the updating embedding. The Tip term anchors predictions to support neighborhoods and language-aligned directions; the CE term exploits encoder plasticity.

Contributions.

1. We locate UniMTS transfer in official cross-dataset few-shot evaluation and show that frozen Tip-style adapters saturate under this protocol.
2. We introduce TipGuided, coupling Tip-cache priors with encoder fine-tuning for motion foundation models.
3. On 18 HAR datasets with official k-shot indices, TipGuided outperforms Tip-F and beats full fine-tuning for all $k \in \{1, 2, 3, 5, 10\}$.

2 Related Work

Motion / IMU foundation models. UniMTS [1] pretrains ST-GCN [4] over a canonical 22-joint layout and aligns embeddings with CLIP text for zero-shot HAR. Our work studies post-hoc few-shot adaptation under UniMTS's public protocol rather than replacing pretraining.

Few-shot adaptation of foundation models. Tip-Adapter and Tip-Adapter-F [2] build keys/values from support features and fuse cache logits with zero-shot CLIP scores. We transplant this mechanism to UniMTS features, then show that a frozen Tip cache is insufficient for cross-dataset IMU shift, motivating encoder-side TipGuided updates. Parameter-efficient fine-tuning (adapters, LoRA) and classical few-shot learners (ProtoNet [5]) share the same budgeted-adaptation goal; TipGuided specifically reuses Tip geometry already native to language-aligned motion encoders.

Fine-tuning under distribution shift. Full fine-tuning remains a strong HAR transfer baseline but erases pretrained structure when supports are tiny. TipGuided is fine-tuning with an explicit nonparametric prior constructed from the same supports.

3 Preliminaries

3.1 UniMTS encoder and text alignment

Let x denote a windowed IMU sequence mapped into a 22-joint skeleton tensor. UniMTS encodes $f(x) \in \mathbb{R}^d$ (d=512) with ST-GCN-18 and compares it to CLIP text embeddings of activity names. We use a four-template prompt ensemble. Zero-shot logits are $z_{\text{text}}(x) = \tau\, f(x)^\top T$ with temperature τ and stacked text matrix T.

3.2 Tip cache

Given a k-shot support set $\{(x_i, y_i)\}_{i=1}^N$, Tip keys are $K = [f(x_i)]$ and values V are class-balanced one-hot labels. For a query with ℓ_2-normalized feature f,

$$A_i = \exp\big(-\beta(1 - \langle f, k_i\rangle)\big), \qquad z_{\text{tip}} = \alpha\, AV.$$

Tip-Adapter-F optimizes keys (and optionally α) on the support set with the encoder frozen.

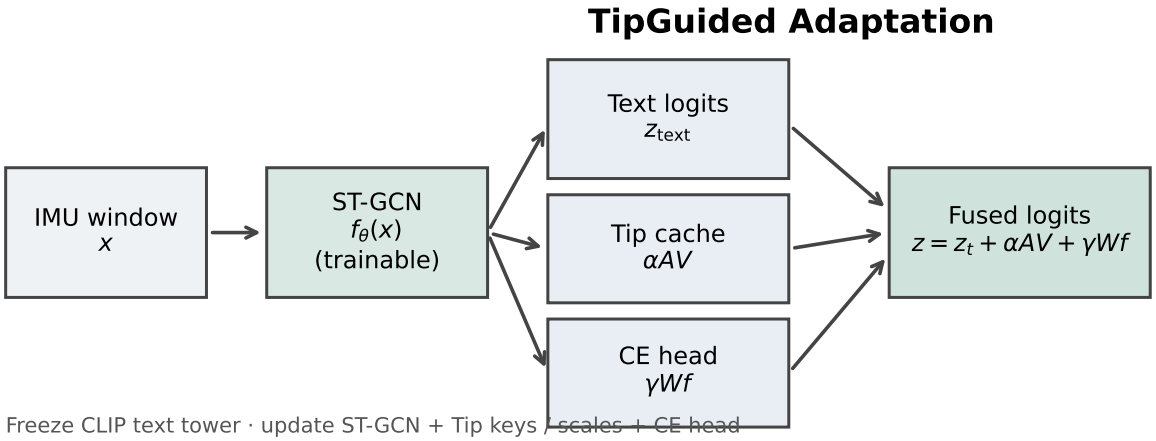


Figure 1: TipGuided. A trainable ST-GCN produces $f_\theta(x)$. Logits fuse text zero-shot scores, Tip-cache affinity over support keys/values, and a CE head. The CLIP text tower remains frozen.

3.3 Evaluation protocol

We adopt the cross-dataset protocol of UniMTS [1]: the encoder is pretrained on simulated motion data and evaluated on 18 real HAR datasets spanning easy / medium / hard activity cardinalities (Table 1). Few-shot supports use the official files `few_shot_data_2/{dataset}_k={k}.pth` with $k \in \{1, 2, 3, 5, 10\}$. For Tip hyper-parameters (β, α), leave-one-dataset-out selection on the other 17 datasets' mean accuracy evaluates the held-out dataset. Metrics follow UniMTS: accuracy and macro-F1, averaged over the 18 datasets. Numbers marked † are transcribed from UniMTS Tables 1–2; all other numbers are from our runs on the same splits.

4 Method: Tip-Guided Adaptation

4.1 Objective

TipGuided initializes from a UniMTS checkpoint, freezes the CLIP text tower, and updates ST-GCN parameters θ together with Tip keys K, scales α, γ, and CE head W. On support samples we minimize cross-entropy of the fused logits in Eq. (1). Keys initialize from current support embeddings and optimize jointly; values remain support-derived and class-balanced.

4.2 What the prior contributes

Full FT optimizes only $W f_\theta(x)$. With tiny N, gradients push f_θ toward support memorization and away from language-aligned geometry. The Tip term keeps predictions consistent with support neighborhoods; z_{text} preserves semantic directions from pretraining. Together they act as a data-dependent regularizer rather than a separate classifier bolted on after adaptation.

4.3 Operating regimes

Ablations identify a clear recipe: **low shot** ($k \leq 3$) favors joint Tip+CE fusion; **mid/high shot** ($k \geq 5$) favors CE-primary training with Tip as initialization/regularizer under moderate learning rates ($\sim 5{\times}10^{-5}$) and 100–150 epochs.

Table 1: Zero-shot averages on 18 datasets † (UniMTS Table 1). Best in **bold**.

Method	Acc (%)	Macro-F1 (%)
ImageBind†	12.5	7.8
IMU2CLIP†	8.6	4.3
IMUGPT†	11.1	5.7
HARGPT†	12.9	6.0
LLaVA†	12.0	3.4
UniMTS†	**43.5**	**34.3**

Table 2: Few-shot averages on 18 datasets (official UniMTS indices; our runs). Acc / Macro-F1 in %. Best per k in **bold**.

k	ZS	1-NN	Probe	Tip-F	Full FT	**TipGuided**
1	44.4/36.0	57.8/54.3	45.8/37.0	59.0/55.3	61.2/53.0	**63.0/57.9**
2	44.4/36.0	63.0/59.9	53.0/44.2	63.2/59.8	67.8/60.4	**71.6/68.2**
3	44.4/36.0	65.0/62.5	54.6/46.1	67.6/65.1	72.8/67.4	**74.6/71.9**
5	44.4/36.0	68.6/66.8	51.9/43.6	70.5/68.0	78.0/72.0	**78.4/73.5**
10	44.4/36.0	70.6/68.7	61.5/52.3	75.1/72.4	81.5/76.9	**83.2/80.5**

5 Experiments

5.1 Baselines

We reuse the UniMTS benchmark methods wherever numerical results are published [1]: ImageBind, IMU2CLIP, IMUGPT, HARGPT, LLaVA (zero-shot); ImageBind, IMU2CLIP, TST, TARNet, TS2Vec, BioBankSSL, THAT, IMUGPT, TimesNet, GPT4TS, SHARE, and Random (full-shot). For few-shot adaptation of UniMTS itself we report our runs: cosine 1-NN, linear probe, Tip-F (LOO), Tip+, Full FT (encoder+CE), and TipGuided. TipGuided uses fuse for $k \leq 3$ and CE-primary for $k \geq 5$.

5.2 Zero-shot context (reported)

Table 1 reproduces UniMTS Table 1 averages. UniMTS zero-shot Acc 43.5 matches our checkpoint evaluation (44.4), confirming protocol alignment before few-shot adaptation.

5.3 Few-shot results

Figure 2 follows the UniMTS Fig. 4 layout: per-dataset Macro-F1 versus number of samples. TipGuided (red) leads on most datasets and budgets. Table 2 reports 18-dataset mean Acc/F1 in the same percentage style as UniMTS. TipGuided is best at every k, gaining up to $+8.4$ Acc points over Tip-F and $+3.8$ over Full FT (k=2).

5.4 Full-shot context (reported)

Table 3 reproduces UniMTS Table 2 averages. Full-shot uses the entire target training split and is an upper reference for the same cross-dataset benchmark; our few-shot TipGuided at k=10 reaches 83.2 Acc, closing much of the gap to reported UniMTS full-shot (90.6).

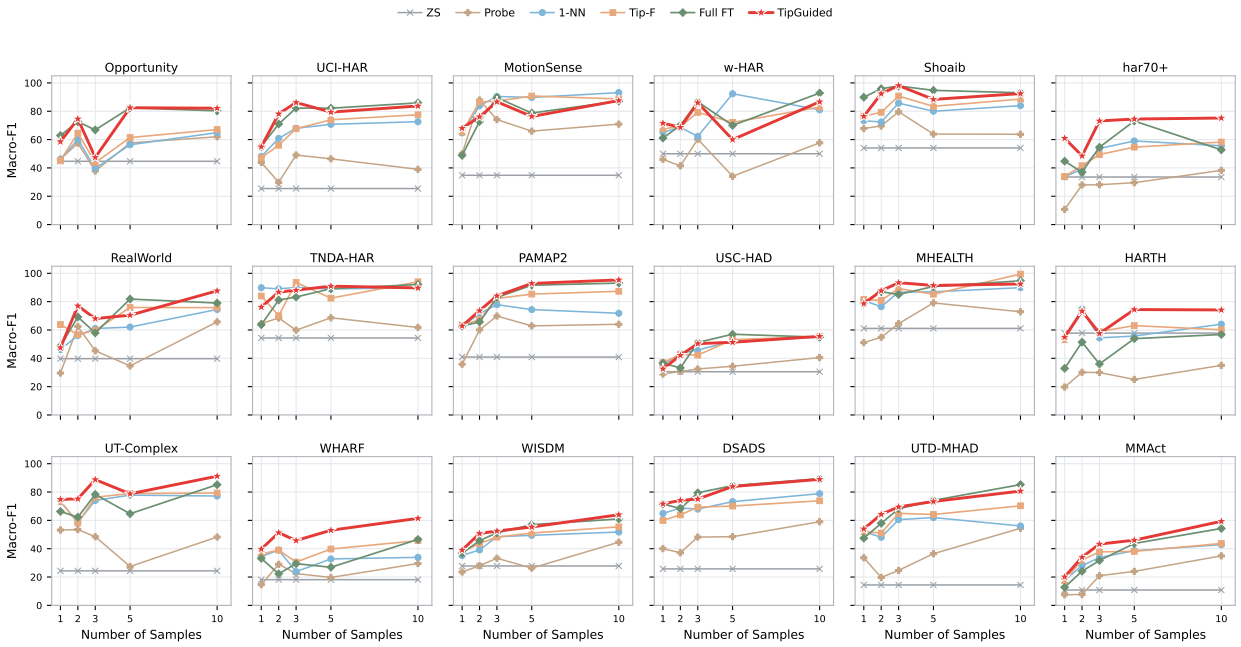


Figure 2: Few-shot fine-tuning results (Macro-F1). Layout mirrors UniMTS Fig. 4 on the same 18 datasets and $k \in \{1, 2, 3, 5, 10\}$. Curves are our runs; TipGuided is highlighted in red.

5.5 Diagnostic ladder

Figure 4a orders methods at k=5. Zero-shot and linear probes sit far below Tip-F, confirming that cache adaptation exploits support geometry beyond a linear head. Tip+ barely moves Tip-F, marking frozen saturation. Full FT jumps into the high 0.78 range; TipGuided clears that bar.

5.6 Per-dataset behavior

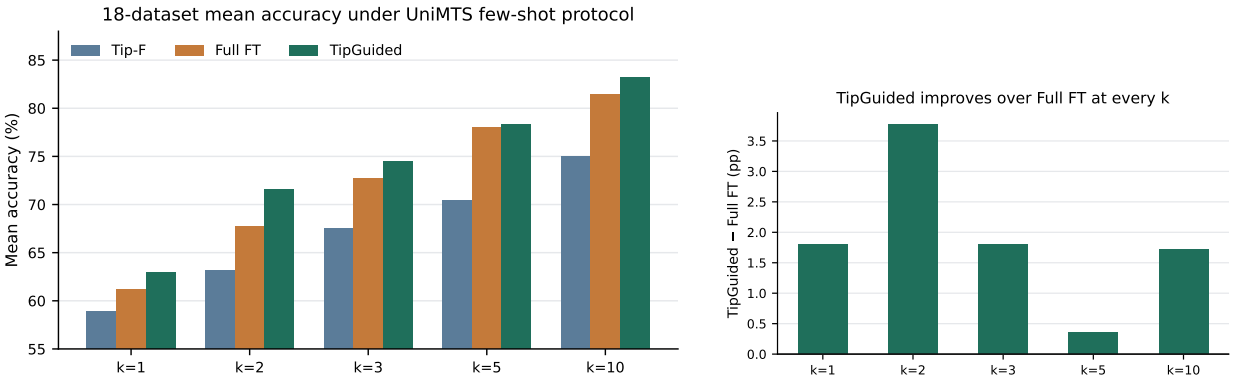
Figure 5 compares Tip-F, Full FT, and TipGuided on each dataset. At k=10, TipGuided wins on $12/18$ datasets, with large gains on w-HAR, Wharf, ut-complex, MMAct, and realworld—settings where prior-free FT overfits heterogeneous or smaller targets. At k=5, TipGuided still trails Full FT on Shoaib, realworld, and USCHAD, while leading on Wharf, UTD-MHAD, and MMAct; the mean remains higher because wins concentrate on hard, low-resource targets.

5.7 Training dynamics

Figure 6 shows Tip-F key training: support CE drops quickly while test accuracy plateaus—the frozen encoder never moves. Figure 7 contrasts Full FT and TipGuided on matched supports and budgets. TipGuided starts from a much stronger test accuracy because Tip/text logits are available from epoch 0, and its fused objective drives support CE down faster than CE-only Full FT. On UCIHAR and PAMAP, TipGuided remains competitive throughout; on Wharf the short-horizon curve is noisier, consistent with that dataset's small test pool. The 18-dataset means in Table 2 aggregate these trajectories under the full training recipe.

5.8 Feature geometry

Figure 8 visualizes t-SNE of test embeddings before and after TipGuided-CE on representative datasets. After adaptation, class clusters tighten and inter-class gaps widen relative to the frozen UniMTS features, matching



(a) Mean accuracy.

(b) Gap vs Full FT.

Figure 3: Aggregate few-shot accuracy. TipGuided exceeds Full FT at every k.

Table 3: Full-shot averages on 18 datasets † (UniMTS Table 2). Best in **bold**.

Method	Acc (%)	Macro-F1 (%)
ImageBind†	85.0	78.1
IMU2CLIP†	80.5	75.0
TST†	79.9	74.2
TARNet†	75.1	65.9
TS2Vec†	84.5	79.8
BioBankSSL†	85.8	82.7
THAT†	80.1	74.5
IMUGPT†	74.4	66.4
TimesNet†	78.0	72.6
GPT4TS†	78.7	72.5
SHARE†	85.8	80.1
Random†	84.7	78.5
UniMTS†	**90.6**	**87.5**

the accuracy gains in Table 2.

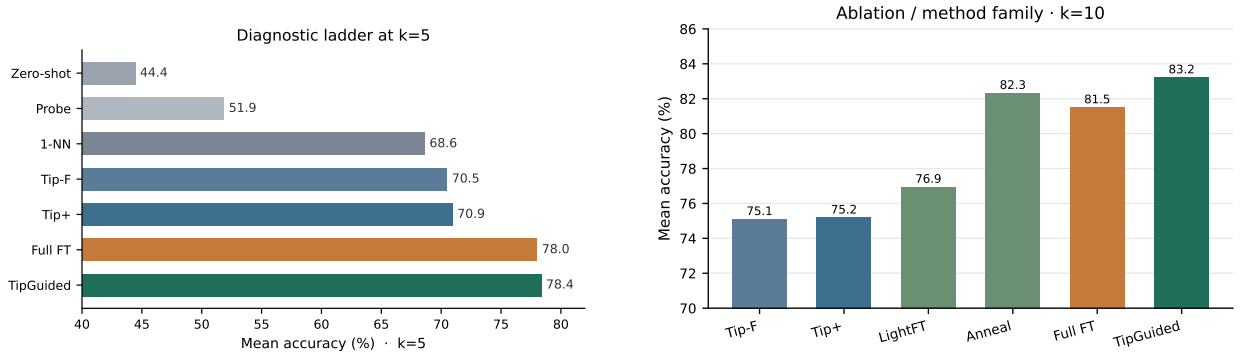
5.9 Ablations

At k=10 (Figure 4b), LightFT and Tip→CE annealing improve over Tip-F yet remain below TipGuided CE (0.832). Longer aggressive schedules without Tip structure do not substitute for the prior: a longer CE-only TipGuided run at k=10 underperforms the default 100-epoch recipe.

6 Discussion

TipGuided reframes Tip-Adapter from a frozen classifier into an adaptation prior for encoder fine-tuning. The central empirical fact is simple: under UniMTS's cross-dataset few-shot protocol, keeping Tip/text geometry while updating the motion encoder beats discarding that geometry (Full FT) and beats refusing to update the encoder (Tip-F).

The k-dependent recipe is itself informative. When supports are extremely sparse, Tip logits supply supervision the CE head cannot yet estimate—hence the large early-epoch gap in Figure 7. When supports



(a) Method ladder at k=5.

(b) Method family at k=10.

Figure 4: Where gains come from. Frozen Tip saturates; encoder updates with Tip priors deliver the remaining lift.

grow, CE-primary decoding dominates and Tip mainly shapes the adaptation trajectory.

Scope. All numbers use UniMTS checkpoints and official splits. Extending TipGuided to other IMU foundation models, confirming the tight k=5 margin with multiple seeds, and collapsing fuse/CE-primary into one tip-weight schedule are concrete follow-ups.

7 Conclusion

Cross-dataset few-shot HAR exposes a clear failure mode of both frozen Tip caches and prior-free full fine-tuning. TipGuided closes that gap by fine-tuning UniMTS under Tip-cache and text priors, improving mean accuracy over Full FT for every official k on 18 datasets. Adaptation should update the encoder while keeping foundation-model geometry explicit.

Acknowledgments

[Anonymous.]

References

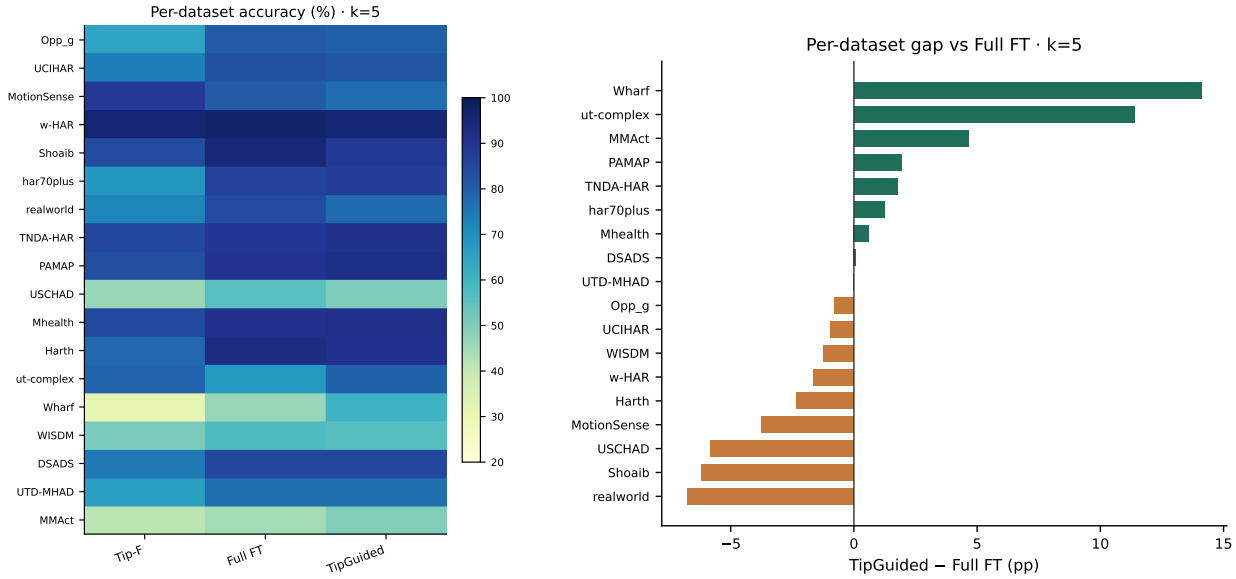
[1] Xiyuan Zhang, Diyan Teng, Ranak Roy Chowdhury, Shuheng Li, Dezhi Hong, Rajesh K. Gupta, and Jingbo Shang. UniMTS: Unified Pre-training for Motion Time Series. In *NeurIPS*, 2024.

[2] Zhang et al. Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification. In *ECCV*, 2022.

[3] Radford et al. Learning Transferable Visual Models From Natural Language Supervision. In *ICML*, 2021.

[4] Yan, Xiong, and Lin. Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition. In *AAAI*, 2018.

[5] Snell, Swersky, and Zemel. Prototypical Networks for Few-shot Learning. In *NeurIPS*, 2017.



(a) Per-dataset accuracy, k=5.

(b) TipGuided − Full FT, k=5.

Figure 5: Dataset-level view. TipGuided concentrates gains where Full FT overfits.

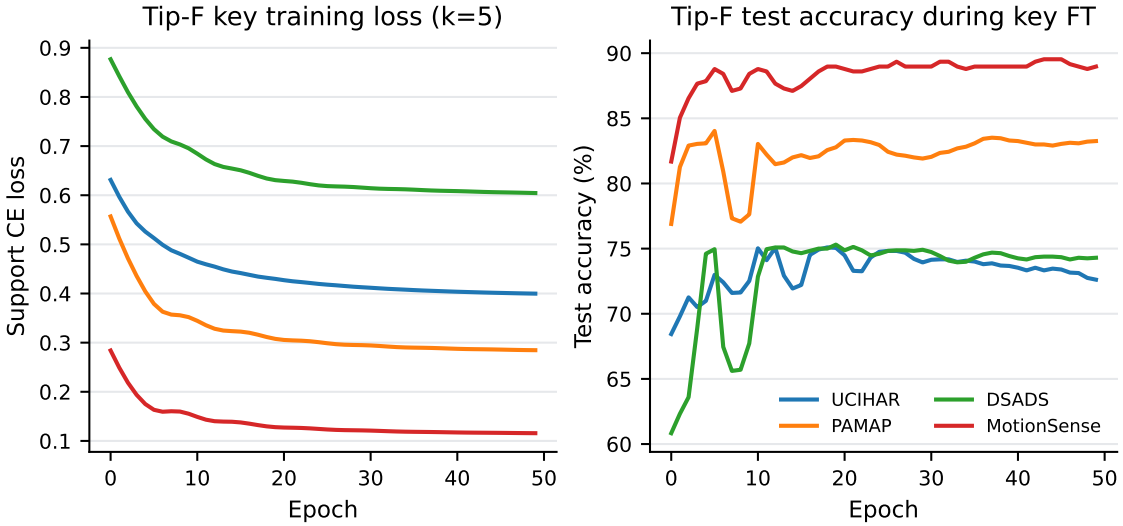


Figure 6: Frozen Tip-F dynamics (k=5). Key fine-tuning fits the support set, yet test accuracy saturates—the encoder never moves.

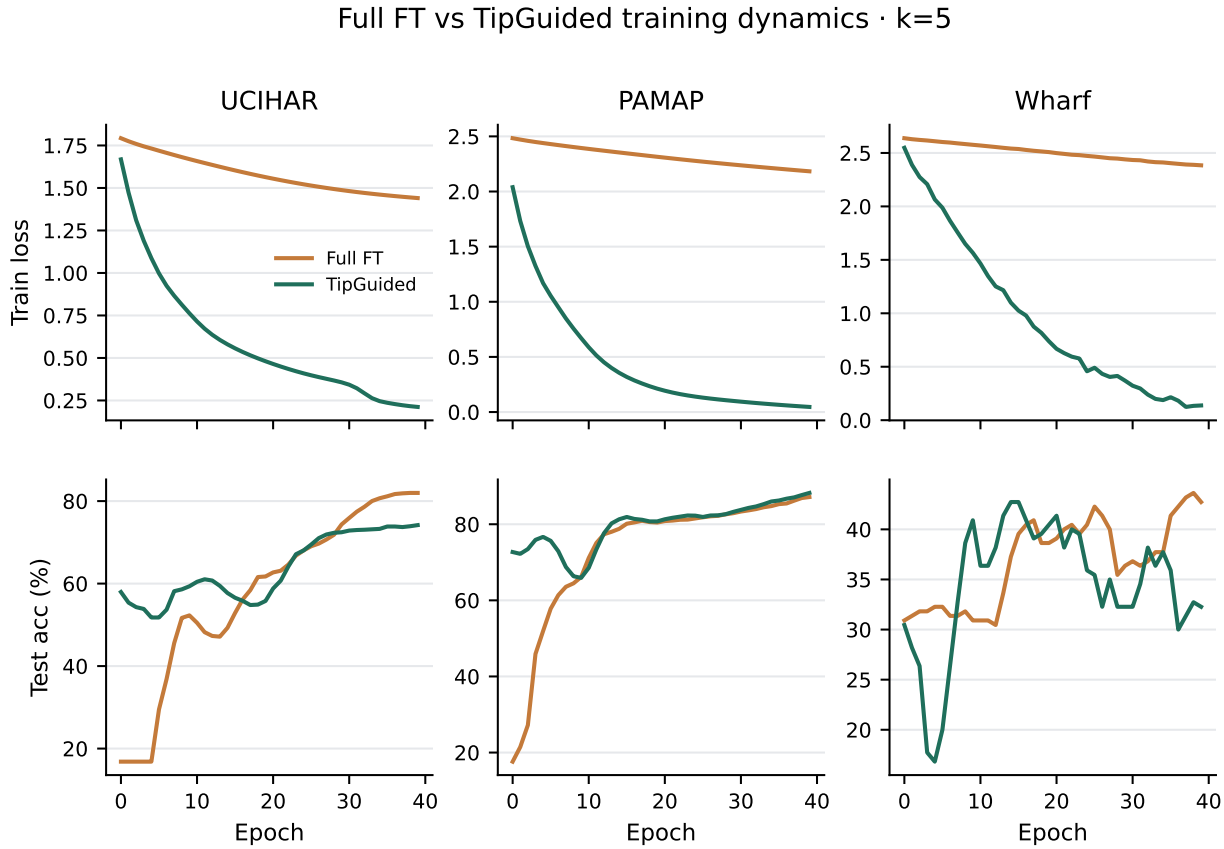


Figure 7: Full FT vs TipGuided training (k=5, 40 epochs). TipGuided enters with higher test accuracy from Tip/text priors and reduces support CE faster via the fused objective.

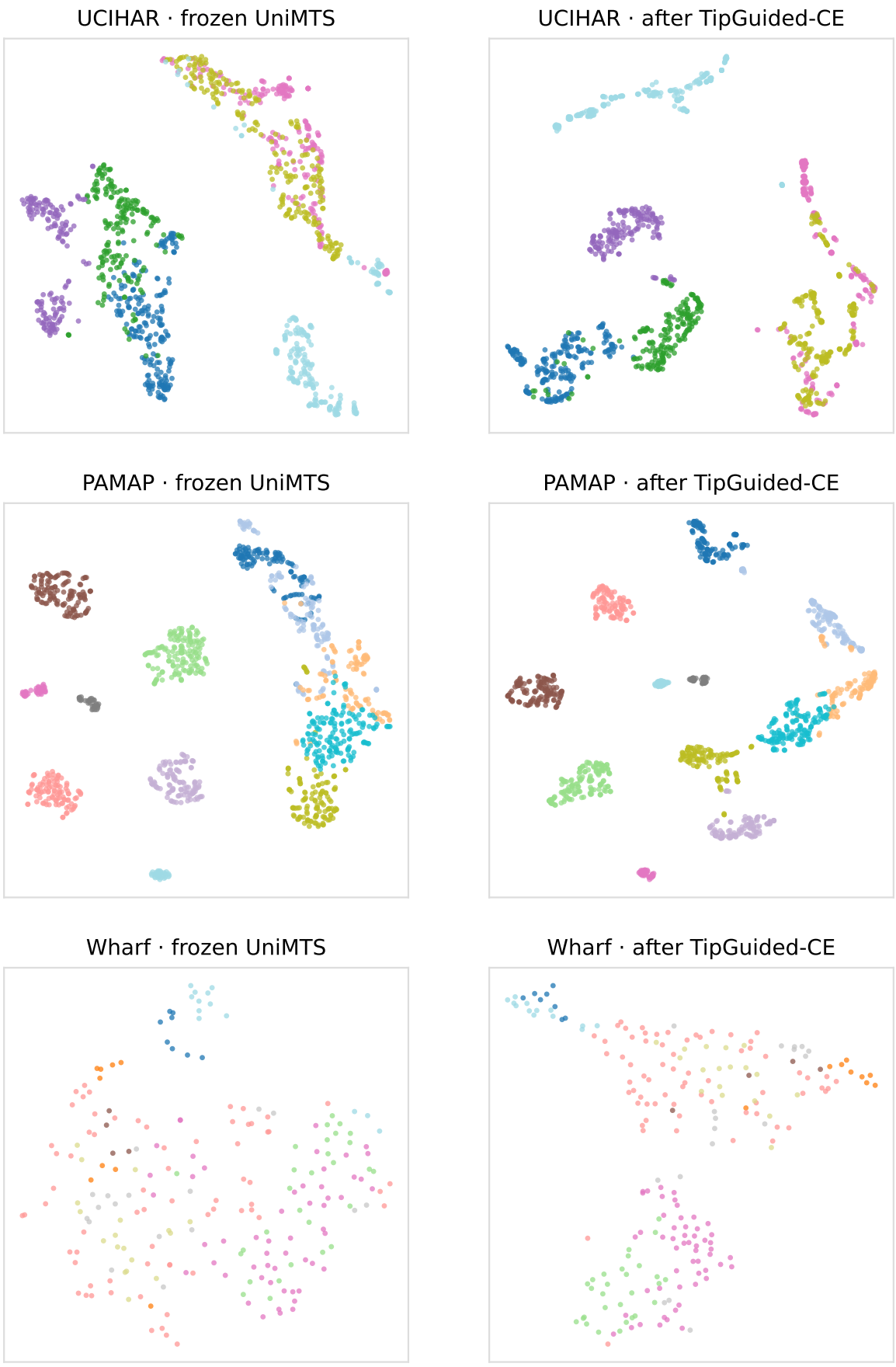


Figure 8: t-SNE of test features (k=5). Each row: frozen UniMTS (left) vs TipGuided-CE (right) for UCIHAR, PAMAP, and Wharf. TipGuided tightens class clusters and widens inter-class gaps.